

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

PHASE 1: Brainstorming & Ideation Phase

This document captures the problem idea, proposed solution, user needs, major modules and the initial concept for PocketSmart AI.

Project Title	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Project Phase	1. Brainstorming & Ideation Phase
Project Domain	Generative AI / Budget Planning / Recommendation System

Backend	FastAPI (project documentation architecture)
Frontend	HTML, CSS, JavaScript / Jinja2 templates

PROJECT TEAM

S.No	Team Member	Role
1	Monish.B	Team Leader
2	S. Jagadeeshwaran	Team Member
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4	Preethika.B	Team Member
5	S. Naresh	Team Member

Document Purpose

This document captures the problem idea, proposed solution, user needs, major modules and the initial concept for PocketSmart AI.

1.1 Project Idea

PocketSmart AI is described as a GenAI-powered cross-platform recommendation system for personalized, budget-based suggestions across home interiors, party planning and jewelry selection. The project documentation describes [LLM](#) for multimodal processing, FastAPI for backend services, HTML/CSS/JavaScript/Jinja2 for the interface, and integrations or simulated sourcing from platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO.

1.2 Problem Identification

- Users often need to manage a fixed budget while comparing many products and services.
- Different planning needs such as home interiors, parties and jewelry require different inputs and recommendation logic.
- Manual comparison across multiple platforms can be time-consuming.
- The project proposes an AI assistant that interprets budget, preferences and context and returns curated recommendations.

1.3 Proposed Solution

- Create one smart assistant with separate planning modules for Home Interior, Party and Jewelry.
- Use [LLM](#) to interpret user inputs and generate contextual recommendations.
- Use optional image input in the Jewelry Planner for outfit-aware suggestions.
- Present recommendations in a structured, user-friendly interface.
- Maintain user session information and recommendation history for personalization and reuse.

1.4 Ideation of Major Modules

Module	Main Inputs	Expected Output
Home Interior Planner	Budget, room type, quantities, preferences	Furniture, decor, lighting and related recommendations
Party Planner	Budget, guest count, event type, venue details	Food, venue, decoration and event recommendations
Jewelry Planner	Budget, occasion, style, optional outfit image	Style-matched jewelry recommendations
Authentication	Registration/login credentials	User session and protected access
History	Past recommendation activity	Saved/reviewable recommendation records

1.5 User Scenarios Considered

- 1 Home interior planning within a defined budget, with room and quantity details.
- 2 Party planning where the system considers budget, guest count and event type.
- 3 Jewelry selection based on budget, occasion, style and optionally an uploaded outfit image.

1.6 Innovation / Value Proposition

- Combines budgeting and recommendation generation in a single workflow.
- Supports multiple lifestyle domains rather than one product category.
- Uses generative AI for contextual responses instead of only fixed recommendation lists.

- Supports multimodal input in the Jewelry Planner.
- Designed around user-specific budget and preference information.

1.7 Initial Feasibility Considerations

- Gemini API access is required for AI generation.
- A web backend is needed to manage requests, authentication, sessions and planner routes.
- External platform data may be sourced through APIs or simulated/mock calls as described in the project documentation.
- Input validation and fallback recommendations are required when AI results are insufficient.

1.8 Ideation Outcome

The brainstorming phase results in a clear concept: PocketSmart AI will function as a budget-aware digital assistant that accepts structured user requirements and produces personalized recommendations for home, party and jewelry planning.